

Machine learning ex1

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1 Question 1

A

$$Cov(x, y) = \mathbb{E}[(X - \mu_x)(Y - \mu_y)] = \sum_x \sum_y P(X = x, Y = y) \cdot (x - \mu_x)(y - \mu_y) \quad (1)$$

since the variables are independent:

$$Cov(x, y) = \sum_x \sum_y P(X = x) \cdot P(Y = y) \cdot (x - \mu_x)(y - \mu_y) = \sum_x P(X = x) \cdot (x - \mu_x) \cdot \sum_y P(Y = y)(y - \mu_y) \quad (2)$$

Now let's look at the following term:

$$\sum_x P(X = x) \cdot (x - \mu_x) = \sum_x P(X = x) \cdot x - \sum_x P(X = x) \cdot \mu_x = \mathbb{E}[X] - \mu_x \cdot \sum_x P(X = x) \quad (3)$$

$$\sum_x P(X = x) \cdot (x - \mu_x) = \mathbb{E}[X] - \mu_x \cdot 1 = \mathbb{E}[X] - \mathbb{E}[X] = 0 \quad (4)$$

Now, from 4,2 we get:

$$Cov(x, y) = \sum_x P(X = x) \cdot (x - \mu_x) \cdot \sum_y P(Y = y)(y - \mu_y) = 0 \cdot \sum_y P(Y = y)(y - \mu_y) = 0 \quad (5)$$

B

Let $X \sim \mathcal{N}(0, 1)$ be a standard normal random variable, and define $Y = X^2$.
It's trivial to see that X, Y are not independent.

Uncorrelated

We compute the covariance between X and Y :

$$Cov(X, Y) = Cov(X, X^2) = \mathbb{E}[X^3] - \mathbb{E}[X] \cdot \mathbb{E}[X^2] \quad (6)$$

Since $X \sim \mathcal{N}(0, 1)$, we have:

$$\mathbb{E}[X] = 0 \quad (7)$$

$$1 = \text{var}(x) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X^2] \quad (8)$$

We also note that:

$$\mathbb{E}[X^3] = 0 \quad (9)$$

Because:

$$\mathbb{E}[X^3] = \int_{-\infty}^{\infty} x^3 \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \quad (10)$$

We denote:

$$f(x) = x^3 \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad (11)$$

$f(x)$ is an **odd function**, namely $f(-x) = -f(x)$, and we are integrating over the symmetric interval $(-\infty, \infty)$. Therefore:

$$\mathbb{E}[X^3] = \int_{-\infty}^{\infty} f(x) dx = 0 \quad (12)$$

So, to sum up the last equations we get

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[X^2] = 1, \quad \mathbb{E}[X^3] = 0 \quad (13)$$

Therefore:

$$\text{Cov}(X, Y) = 0 - 0 \cdot 1 = 0 \quad (14)$$

Hence, X and Y are **uncorrelated**.

C).

let's define X, Y, Z such that X, Y are independent and:

$$P(X = 0) = P(X = 1) = 0.5, P(Y = 0) = P(Y = 1) = 0.5, Z = \text{XOR}(X, Y)$$

It is easy to see by the definition of XOR that:

$$P(Z = 0) = P(Z = 1) = 0.5$$

Now let's show that X, Z are independent and a similar proof will indicate the same for Y, Z .

$$P(X = 0, Z = 0) = P(Z = 0 | X = 0) \cdot P(X = 0)$$

By definition of XOR:

$$P(X = 0, Z = 0) = P(Y = 1) \cdot 0.5 = 0.5 \cdot 0.5 = 0.25 = P(X = 0) \cdot P(Z = 0)$$

And identical proofs can be shown for

$$P(X = 1, Z = 0), P(X = 0, Z = 1), P(X = 1, Z = 1)$$

It is because for each possible value of X there is exactly one possible value of Y that leads to each possible value of Z

D

let there be some column vector z of size d .

We start by expanding the expression:

$$z^\top C z = z^\top \left(\sum_{i=1}^n x_i x_i^\top \right) z = \sum_{i=1}^n z^\top x_i x_i^\top z$$

Note that $z^\top x_i$ is a scalar, so we can write:

$$z^\top x_i x_i^\top z = (z^\top x_i)(x_i^\top z) = (x_i^\top z)^2 \geq 0$$

Therefore,

$$z^\top C z = \sum_{i=1}^n (x_i^\top z)^2 \geq 0$$

Each term in the summation is non-negative, so the whole sum is non-negative.

Q2

A

We know that $X \sim \mathcal{N}(0, 1)$

$P(\text{a sample is within one standard deviation from the mean}) = P(-1 \leq X \leq 1)$

$$P(-1 \leq X \leq 1) = \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \phi(1) - \phi(-1) \approx 0.68$$

B

We want to show that $P(\|X\| \leq 1) \xrightarrow{d \rightarrow \infty} 0$

$$P(\|X\| \leq 1) = P\left(\sqrt{\sum_{i=1}^d x_i^2} \leq 1\right)$$

Lemma: Let's prove that $\|X\| \leq 1 \rightarrow \forall 1 \leq i \leq d : -1 \leq x_i \leq 1$

lets assume for contradiction that there exist some j for which $\|x_j\| > 1$ then

$$\|X\| = \sqrt{\sum_{i=1}^d x_i^2} \geq \sqrt{x_j^2} = \|x_j\| > 1$$

And we get a contradiction to the assumption.

From the Lemma, we get:

$$P(\|X\| \leq 1) = P\left(\sqrt{\sum_{i=1}^d x_i^2} \leq 1\right) \leq P\left(\bigcap_{i=1}^d -1 \leq x_i \leq 1\right)$$

We saw in class that if the covariance matrix is diagonal, then the variables are independent, so:

$$P(\|X\| \leq 1) \leq P\left(\bigcap_{i=1}^d -1 \leq x_i \leq 1\right) = \prod_{i=1}^d P(-1 \leq x_i \leq 1)$$

From section A of this question we get

$$0 \leq P(\|X\| \leq 1) \leq \prod_{i=1}^d P(-1 \leq x_i \leq 1) = (\phi(1) - \phi(-1))^d \approx (0.68)^d \xrightarrow{d \rightarrow \infty} 0$$

Therefore

$$P(\|X\| \leq 1) \xrightarrow{d \rightarrow \infty} 0$$

Q3

The variance of y is:

$$\text{Var}(y) = \mathbb{E}[(y - \mathbb{E}[y])^2] = \mathbb{E}[(w^\top X - \mathbb{E}[w^\top X])^2]$$

Since expectation is linear, we have:

$$\mathbb{E}[w^\top X] = w^\top \mathbb{E}[X]$$

Therefore,

$$\text{Var}(y) = \mathbb{E}[(w^\top X - w^\top \mathbb{E}[X])^2] = \mathbb{E}[(w^\top (X - \mathbb{E}[X]))^2]$$

$$\text{Var}(y) = \mathbb{E}[(w^\top (X - \mathbb{E}[X]))(w^\top (X - \mathbb{E}[X]))^\top] = \mathbb{E}[w^\top (X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top w]$$

Since w is not random, we can take it outside the expectation:

$$\text{Var}(y) = w^\top \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top] w = w^\top C w$$

Q4

A

Let λ be an eigenvalue of A , with a corresponding eigenvector $v \neq 0$, such that:

$$Av = \lambda v$$

Recall that the characteristic polynomial of a matrix is given by:

$$p_A(\lambda) = \det(A - \lambda I)$$

Note that the determinant of a matrix is equal to the determinant of its transpose:

$$\det(A - \lambda I) = \det((A - \lambda I)^\top) = \det(A^\top - \lambda I)$$

Therefore, the characteristic polynomials of A and $A^\top$ are the same:

$$p_A(\lambda) = p_{A^\top}(\lambda)$$

Since the eigenvalues of a matrix are the roots of its characteristic polynomial, A and $A^\top$ must have the same eigenvalues.

B

If A is diagonalizable, then we saw that there exists an invertible matrix P and a diagonal matrix D such that:

$$A = PDP^{-1}$$

where $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ and each λ_i is an eigenvalue of A .

Using the multiplicative property of determinants, we compute:

$$\det(A) = \det(PDP^{-1}) = \det(P) \det(D) \det(P^{-1})$$

Since $\det(P^{-1}) = \frac{1}{\det(P)}$, we get:

$$\det(A) = \det(P) \det(D) \cdot \frac{1}{\det(P)} = \det(D)$$

Finally, since D is diagonal with eigenvalues $\lambda_1, \dots, \lambda_n$ on its diagonal, we have:

$$\det(A) = \det(D) = \prod_{i=1}^n \lambda_i$$

Q5

A

Differentiability

This function is differentiable everywhere except at the boundary $w^\top x = 1$, where the function transitions between the two cases, because, as we will see in the next section, the gradient changes from 0 to $-w$. Therefore, $f(w, x)$ is differentiable in the regions:

$$w^\top x > 1, w^\top x < 1$$

Gradient Computation

- If $w^\top x \geq 1$, then $f(w, x) = 0$ and the gradient is:

$$\nabla f(w, x) = 0.$$

- If $w^\top x < 1$, then $f(w, x) = 1 - w^\top x = 1 - \sum_{i=1}^d x_i \cdot w_i$, so:

$$\nabla f(w, x) = -w.$$

Therefore:

$$\nabla f(w, x) = \begin{cases} 0 & \text{if } w^\top x \geq 1 \\ -w & \text{if } w^\top x < 1 \end{cases}$$

B

Differentiability

The sigmoid function is differentiable everywhere in $\mathbb{R}^d$.

Gradient Computation

Let $f(x) = \sigma(w^\top x)$ where $\sigma(z) = \frac{1}{1+e^{-z}}$.

By the chain rule:

$$\sigma'(z) = \frac{e^{-z}}{(e^{-z} + 1)^2} = \frac{1}{(e^{-z} + 1)} \cdot \frac{e^{-z} + 1 - 1}{(e^{-z} + 1)} = \sigma(z)(1 - \sigma(z))$$

Since $z = w^\top x$, the gradient with respect to x is:

$$\nabla f(x) = \sigma(w^\top x)(1 - \sigma(w^\top x)) \cdot w = f(x)(1 - f(x))w$$